

Project Report: End of the Experiment

Group Members:

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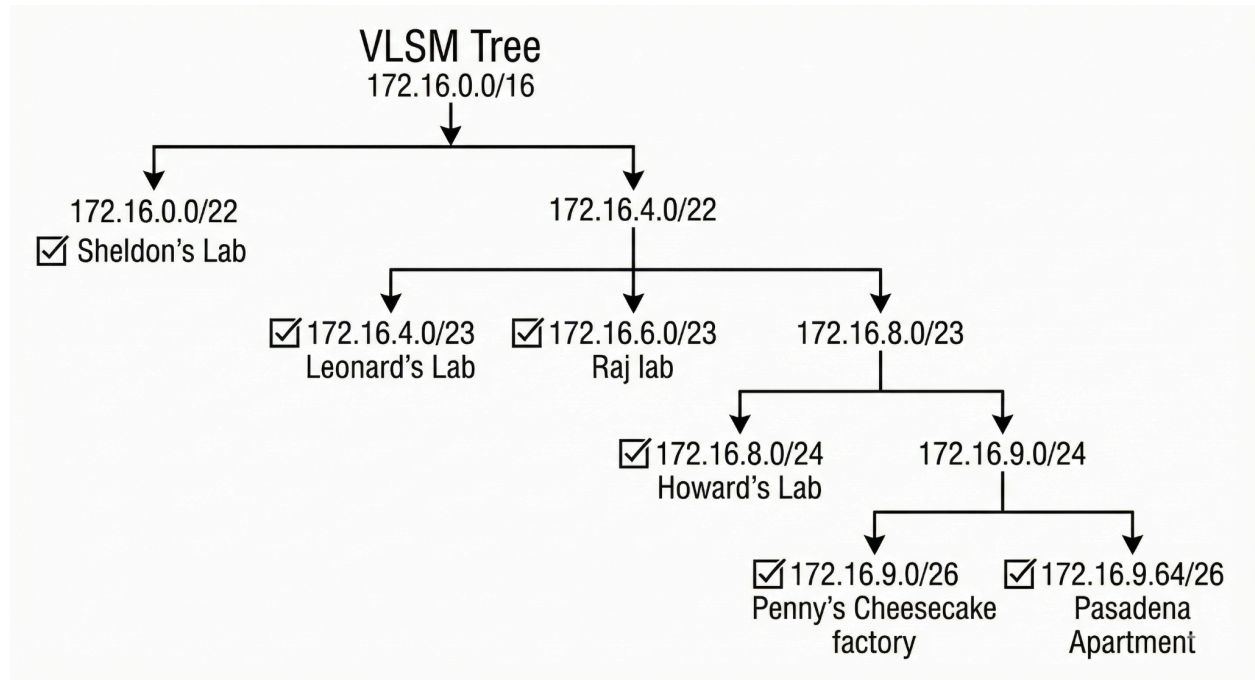
1. VLSM Tree

The base network 172.16.0.0/16 was divided based on the host requirements for each department/zone.

- **Root Network:** 172.16.0.0/16
- **Sheldon's Lab (812 Hosts):** 172.16.0.0/22 (Range: 172.16.0.1 – 172.16.3.254)
- **Leonard's Lab (360 Hosts):** 172.16.4.0/23 (Range: 172.16.4.1 – 172.16.5.254)
- **Raj's Lab (275 Hosts):** 172.16.6.0/23 (Range: 172.16.6.1 – 172.16.7.254)
- **Howard's Lab (190 Hosts):** 172.16.8.0/24 (Range: 172.16.8.1 – 172.16.8.254)
- **Penny's Cheesecake Factory (58 Hosts):** 172.16.9.0/26 (Range: 172.16.9.1 – 172.16.9.62)
- **Pasadena apartment (60 Hosts):** 172.16.9.64/26 (Range: 172.16.9.65 – 172.16.9.126)

WAN Links (/30 subnets):

- **Link 1 (RB ↔ RC):** 172.16.9.128/30 (IPs: 172.16.9.129, 172.16.9.130)
- **Link 2 (RB ↔ RA):** 172.16.9.132/30 (IPs: 172.16.9.133, 172.16.9.134)
- **Link 3 (RA ↔ RC):** 172.16.9.136/30 (IPs: 172.16.9.137, 172.16.9.138)



2. Address Table

Device Name	Interface	IP Address	Subnet Mask	Default Gateway	Description
Router-C	G0/0/0	172.16.4.1	255.255.254.0	N/A	Leonard's Gateway
	G0/0/1	172.16.8.1	255.255.255.0	N/A	Howard's Gateway
	S0/1/0	172.16.9.130	255.255.255.252	N/A	Link to Router-B
	S0/1/1	172.16.9.138	255.255.255.252	N/A	Link to Router-A
Router-B	G0/0/0	172.16.0.1	255.255.252.0	N/A	Sheldon's Gateway
	G0/0/1	172.16.9.1	255.255.255.192	N/A	Penny's Gateway

	S0/1/1	172.16.9.133	255.255.255.252	N/A	Link to Router-A
	S0/1/0	172.16.9.129	255.255.255.252	N/A	Link to Router-C
Router-A	G0/0/0	172.16.6.1	255.255.254.0	N/A	Raj's Gateway
	G0/0/1	172.16.9.65	255.255.255.192	N/A	Apartment Gateway
	S0/1/0	172.16.9.134	255.255.255.252	N/A	Link to Router-B
	S0/1/1	172.16.9.137	255.255.255.252	N/A	Link to Router-C
Sheldon Svr	NIC	172.16.0.2	255.255.252.0	172.16.0.1	Email Server
Raj Svr	NIC	172.16.6.2	255.255.254.0	172.16.6.1	Email Server
Apt Svr	NIC	172.16.9.66	255.255.255.192	172.16.9.65	DNS / Web Server
Apt PC	NIC	172.16.9.67	255.255.255.192	172.16.9.65	Static Client

3. Configuration Commands

Router A Configuration (Raj & Apartment)

```
Router> en
Router# config t
Router(config)# hostname Router-A
```

[Interface: Raj's Lab]

```
Router-A(config)# interface GigabitEthernet0/0/0
Router-A(config-if)# ip address 172.16.6.1 255.255.254.0
Router-A(config-if)# no shutdown
Router-A(config-if)# exit
```

[Interface: Apartment Committee]

```
Router-A(config)# interface GigabitEthernet0/0/1
Router-A(config-if)# ip address 172.16.9.65 255.255.255.192
```

```
Router-A(config-if)# no shutdown
Router-A(config-if)# exit
```

[Interface: Serial Link to Router-B]

```
Router-A(config)# interface Serial0/1/1
Router-A(config-if)# ip address 172.16.9.134 255.255.255.252
Router-A(config-if)# clock rate 64000
Router-A(config-if)# no shutdown
Router-A(config-if)# exit
```

[Interface: Serial Link to Router-C]

```
Router-A(config)# interface Serial0/1/0
Router-A(config-if)# ip address 172.16.9.137 255.255.255.252
Router-A(config-if)# clock rate 64000
Router-A(config-if)# no shutdown
Router-A(config-if)# exit
```

[DHCP Configuration]

```
Router-A(config)# ip dhcp pool RAJ
Router-A(dhcp-config)# network 172.16.6.0 255.255.254.0
Router-A(dhcp-config)# default-router 172.16.6.1
Router-A(dhcp-config)# dns-server 172.16.9.66
Router-A(dhcp-config)# exit
```

[Routing Protocol (RIPv2)]

```
Router-A(config)# router rip
Router-A(config-router)# version 2
Router-A(config-router)# no auto-summary
Router-A(config-router)# network 172.16.0.0
Router-A(config-router)# exit
```

[Static Route to Leonard's Lab (via Router-C)]

```
Router-A(config)# enable
Router-A(config)# config t
Router-A(config)# ip route 172.16.4.0 255.255.254.0 172.16.9.138
```

[Static Route to Penny's Lab (via Router-B)]

```
Router-A(config)# ip route 172.16.9.0 255.255.255.192 172.16.9.133
Router-A(config)# exit
```

[Floating Static Route to Raj's Lab via Router-C (Backup)]

```
Router-A(config)# ip route 172.16.6.0 255.255.254.0 172.16.9.138 25
Router-A(config)# exit
```

Router B Configuration (Sheldon & Penny)

```
Router> en
Router# config t
Router(config)# hostname Router-B
```

[Interface: Sheldon's Lab]

```
Router-B(config)# interface GigabitEthernet0/0/0
Router-B(config-if)# ip address 172.16.0.1 255.255.252.0
Router-B(config-if)# no shutdown
Router-B(config-if)# exit
```

[Interface: Penny's Network]

```
Router-B(config)# interface GigabitEthernet0/0/1
Router-B(config-if)# ip address 172.16.9.1 255.255.255.192
Router-B(config-if)# no shutdown
Router-B(config-if)# exit
```

[Interface: Serial Link to Router-C]

```
Router-B(config)# interface Serial0/1/1
Router-B(config-if)# ip address 172.16.9.129 255.255.255.252
Router-B(config-if)# clock rate 64000
Router-B(config-if)# no shutdown
Router-B(config-if)# exit
```

[Interface: Serial Link to Router-A]

```
Router-B(config)# interface Serial0/1/0
Router-B(config-if)# ip address 172.16.9.133 255.255.255.252
Router-B(config-if)# clock rate 64000
Router-B(config-if)# no shutdown
Router-B(config-if)# exit
```

[DHCP Configuration]

```
Router-B(config)# ip dhcp pool SHELDON
Router-B(dhcp-config)# network 172.16.0.0 255.255.252.0
Router-B(dhcp-config)# default-router 172.16.0.1
Router-B(dhcp-config)# dns-server 172.16.9.66
Router-B(dhcp-config)# exit
```

```
Router-B(config)# ip dhcp pool PENNY
Router-B(dhcp-config)# network 172.16.9.0 255.255.255.192
Router-B(dhcp-config)# default-router 172.16.9.1
Router-B(dhcp-config)# dns-server 172.16.9.66
Router-B(dhcp-config)# exit
```

[Routing Protocol (RIPv2)]

```
Router-B(config)# router rip
```

```
Router-B(config-router)# version 2
Router-B(config-router)# no auto-summary
Router-B(config-router)# network 172.16.0.0
Router-B(config-router)# exit
```

[Static Route to Leonard's Lab (via Router-C)]

```
Router-B(config)# enable
Router-B(config)# config t
Router-B(config)# ip route 172.16.4.0 255.255.254.0 172.16.9.130
Router-B(config)# exit
```

Router-C Configuration (Leonard & Howard)

```
Router> en
Router# config t
Router(config)# hostname Router-C
```

[Interface: Leonard's Lab]

```
Router-C(config)# interface GigabitEthernet0/0/0
Router-C(config-if)# ip address 172.16.4.1 255.255.254.0
Router-C(config-if)# no shutdown
Router-C(config-if)# exit
```

[Interface: Howard's Lab]

```
Router-C(config)# interface GigabitEthernet0/0/1
Router-C(config-if)# ip address 172.16.8.1 255.255.255.0
Router-C(config-if)# no shutdown
Router-C(config-if)# exit
```

[Interface: Serial Link to Router-B]

```
Router-C(config)# interface Serial0/1/1
Router-C(config-if)# ip address 172.16.9.130 255.255.255.252
Router-C(config-if)# clock rate 64000
Router-C(config-if)# no shutdown
Router-C(config-if)# exit
```

[Interface: Serial Link to Router-A]

```
Router-C(config)# interface Serial0/1/0
Router-C(config-if)# ip address 172.16.9.138 255.255.255.252
Router-C(config-if)# clock rate 64000
Router-C(config-if)# no shutdown
Router-C(config-if)# exit
```

[DHCP Configuration]

```
Router-C(config)# ip dhcp pool LEONARD
```

```
Router-C(dhcp-config)# network 172.16.4.0 255.255.254.0
Router-C(dhcp-config)# default-router 172.16.4.1
Router-C(dhcp-config)# dns-server 172.16.9.66
Router-C(dhcp-config)# exit
```

```
Router-C(config)# ip dhcp pool HOWARD
Router-C(dhcp-config)# network 172.16.8.0 255.255.255.0
Router-C(dhcp-config)# default-router 172.16.8.1
Router-C(dhcp-config)# dns-server 172.16.9.66
Router-C(dhcp-config)# exit
```

[Routing Protocol (RIPv2)]

```
Router-C(config)# router rip
Router-C(config-router)# version 2
Router-C(config-router)# no auto-summary
Router-C(config-router)# network 172.16.0.0
Router-C(config-router)# exit
```

[Static Route to Penny's Lab (via Router-B)]

```
Router-C(config)# enable
Router-C(config)# config t
Router-C(config)# ip route 172.16.9.0 255.255.255.192 172.16.9.129
Router-C(config)# exit
```

Server Configurations

1. Apartment Server (Web & DNS)

- **IP Configuration:**
 - Static IP 172.16.9.66
 - **Mask:** 255.255.255.192
 - **Gateway:** 172.16.9.65
- **HTTP:** On. Displayed "Bazinga! We've got your back!".
- **DNS:** On.
 - Record 1: sheldon.com -> 172.16.0.2
 - Record 2: raj.com -> 172.16.6.2
 - Record 3: www.caltechsupport.com -> 172.16.9.66

2. Sheldon's Email Server

- **IP Configuration:**
 - Static IP 172.16.0.2
 - Mask 255.255.252.0

- Gateway 172.16.0.1
 - DNS 172.16.9.66
- **Email:** On Domain: sheldon.com. User: sheldon, Pass: 74328

3. Raj's Email Server

- **IP Configuration:**
 - Static IP 172.16.6.2
 - Mask 255.255.254.0
 - Gateway 172.16.6.1
 - DNS 172.16.9.66
- **Email:** On. Domain: raj.com. User: raj, Pass: 74328

4. List of Assumptions Made

- We assumed /30 subnets for all router-to-router links to conserve IP addresses.
- We assumed a triangular mesh topology. Traffic reroutes via neighbor routers if a direct link fails.
- A single DNS server (172.16.9.66) serves all labs.
- Cheesecake factory(Penny) & Experimental Physics Lab(Leonard) is set to static routing. Rest of others are dynamic routing.
- We set 60 hosts for Pasadena Apartment Committee. [As the host number is not mentioned]